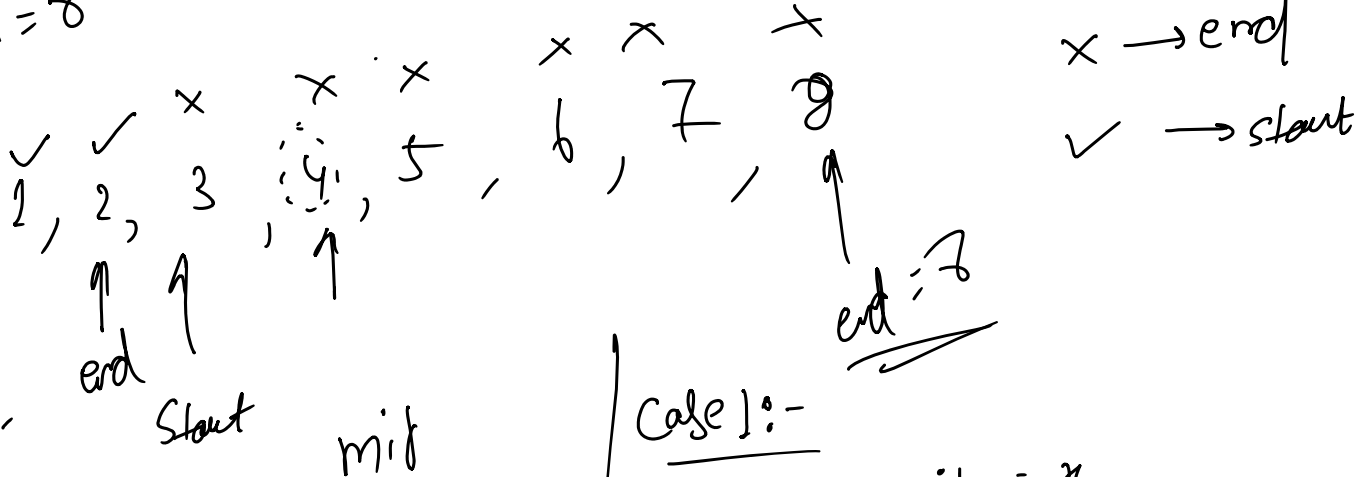


finding Square root using Binary Search;

$n=8$



$$\rightarrow Sq = 2 \times 2$$

$$Sq = 3 \times 3$$

ans $\rightarrow$ last ✓.

Case 1:-

$$mid \times mid == x$$

$\rightarrow$ return mid

Case 2:- $mid \times mid > x$
 $e = mid - 1;$

Case 3:- $mid \times mid < x$

$$s = mid + 1$$

return end;

$$\frac{start + end}{2}$$

// recursive

$$\text{Start} + \frac{(\text{end} - \text{start})}{2}$$

$$\downarrow \quad \downarrow$$
$$\text{Start} + \frac{\text{end}}{2} - \frac{\text{Start}}{2}$$

$$\frac{\text{Start}}{2} + \frac{\text{end}}{2} = \frac{(\text{Start} + \text{end})}{2}$$

n^{th} root of m

$$m = 8$$

$$n = 3$$

$$\sqrt[3]{8}$$

$$m = 9$$

$$n = 2$$

$$\sqrt{9} = 3$$

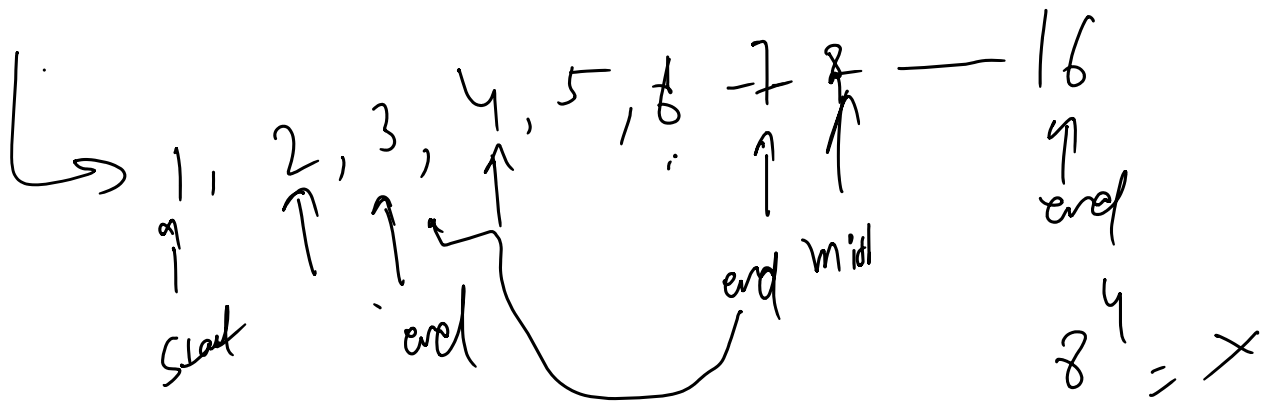
$$m = 16$$
$$h = 4$$

$$\sqrt[4]{16} = 2$$

Ans.

n^{th} root of m

$$m = 16, \underline{n = 4}$$



$$2^4 = 2 \checkmark$$

return mid⁴ x